

PARANG, Romania

WMO: 153200

Lat: 45.3875N

Lon: 23.4631E

Elev: 1549

StdP: 84.05

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-13.4	-20.9	0.7	-6.8	-17.5	1.0	-9.1	9.9	-6.1	8.1	-4.5	1.0	0	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	21.9	15.5	20.5	14.9	19.3	14.4	16.8	20.2	16.0	19.3	15.2	18.4	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	13.5	18.4	14.7	12.6	17.6	13.9	12.0	16.9	53.7	20.4	50.7	19.4	48.2	18.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.2	4.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-4.8	-3.0	-1.0	3.7	8.7	12.8	13.9	14.6	9.6	5.4	2.2	-2.7
	DBStd	7.96	4.99	4.53	4.87	5.39	3.51	3.43	3.16	3.24	4.53	4.95	4.59	4.56
	HDD10.0	2299	458	363	343	198	65	14	6	4	64	157	236	393
	HDD18.3	4882	716	596	601	440	299	168	142	121	263	400	484	652
	CDD10.0	469	0	0	1	8	24	97	126	146	52	16	1	0
	CDD18.3	10	0	0	0	0	0	1	3	5	1	0	0	0
	CDH23.3	13	0	0	0	0	0	1	4	7	2	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	1.5	1.3	1.6	1.5	1.7	1.4	1.5	1.6	1.4	1.5	1.4	1.8	1.4
Precipitation	PrecAvg	794	41	38	45	66	89	109	110	67	74	58	52	50
	PrecMax	1105	83	97	92	122	166	203	221	132	149	139	165	108
	PrecMin	496	10	6	14	16	30	24	41	14	5	1	0	8
	PrecStd	157	20	21	24	27	37	46	49	31	37	32	36	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.0	12.0	17.2	18.5	22.1	23.2	24.0	22.6	17.9	14.1	10.9
		MCWB	1.1	4.8	5.3	9.2	12.6	15.5	15.7	16.1	15.9	9.7	6.2	2.6
	2%	DB	4.6	6.4	9.4	15.0	16.8	20.5	21.3	22.2	19.4	16.1	11.7	6.9
		MCWB	0.7	3.1	4.4	9.0	11.8	15.8	15.4	15.8	12.8	9.8	5.9	1.9
	5%	DB	3.0	4.3	7.3	12.9	15.2	19.2	20.0	20.6	17.3	14.0	9.9	4.7
		MCWB	0.5	1.5	2.9	7.5	11.1	15.2	15.3	14.6	12.5	8.7	5.4	1.7
	10%	DB	1.7	2.8	5.5	10.8	13.8	17.9	18.6	19.2	15.7	12.3	8.2	2.9
		MCWB	-0.4	0.9	2.3	6.4	10.2	14.3	14.4	14.0	11.5	7.8	5.0	0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	5.7	6.5	10.8	14.3	18.0	17.6	17.8	15.7	12.4	8.9	4.7
		MCDB	5.1	7.6	10.7	15.5	17.7	20.3	20.8	22.1	20.7	14.7	12.7	7.1
	2%	WB	2.3	3.8	4.8	9.2	12.6	16.7	16.6	16.5	14.1	10.7	7.6	3.1
		MCDB	3.7	5.9	8.3	14.1	15.7	19.4	19.8	20.4	18.0	14.0	10.0	5.8
	5%	WB	1.2	2.3	3.7	8.1	11.5	15.7	15.7	15.6	13.0	9.7	6.8	1.9
		MCDB	2.6	4.0	6.5	12.0	14.6	18.8	18.9	19.0	16.6	13.1	8.7	4.2
	10%	WB	0.1	1.2	2.5	7.0	10.5	14.6	15.0	14.9	11.9	8.7	5.9	1.1
		MCDB	1.3	2.4	5.0	10.8	13.4	17.5	18.0	18.2	14.9	11.7	7.5	2.5
Mean Daily Temperature Range	5% DB	MDBR	6.1	5.9	6.2	6.6	6.2	6.6	6.8	6.4	5.9	5.7	5.3	5.3
		MCDBR	8.0	7.0	8.1	8.6	8.0	8.0	8.1	7.4	7.3	7.6	6.8	6.8
		MCWBR	4.9	4.7	4.8	5.2	5.2	5.4	5.1	4.6	4.6	5.6	4.2	4.2
	5% WB	MCDBR	6.1	6.2	7.2	8.2	7.5	7.8	7.3	6.8	6.9	6.7	5.3	5.9
		MCWBR	4.5	4.6	4.9	5.1	5.1	5.6	5.1	4.8	4.7	5.4	3.6	4.4
Clear-Sky Solar Irradiance	taub	0.262	0.286	0.320	0.368	0.357	0.355	0.368	0.379	0.344	0.314	0.292	0.265	
	taud	2.411	2.358	2.356	2.276	2.351	2.378	2.337	2.304	2.391	2.484	2.498	2.454	
	Ebn at Noon	863	900	912	890	910	911	893	869	874	859	820	823	
	Edh at Noon	71	91	106	125	120	118	121	121	101	80	66	61	
All-Sky Solar Radiation	RadAvg	1.33	2.13	3.24	4.37	5.21	5.86	5.78	5.24	3.74	2.60	1.53	1.09	
	RadStd	0.17	0.28	0.45	0.48	0.47	0.55	0.47	0.51	0.38	0.30	0.19	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)		+0.64	N/A	N/A	+0.89	N/A	N/A	-114	-168	+91	+44

Nomenclature: See separate page